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Title: Assembly Line Bill of Process (BOP)	
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1.0 INTRODUCTION

The Bill of Process (BoP) defines the required attributes of a best-in-class process. The material presented in this document was generated and is maintained by the Advanced Manufacturing Engineering team to document key process learnings and communicate expectations for assembly line design and operation. This Assembly Bill of Process describes the critical attributes to include when designing a new assembly Line or improving an existing one. The basis for comparing competing methods is impact to Safety, Quality, Delivery, and Cost (in this order) and the philosophy behind line design and operation is TPS (Toyota Production System) or Lean rules and ideals. This Bill of Process (BoP) is a standard for achieving predictable, best in class process performance.

2.0 SCOPE

The scope of this document is limited to assembly line manufacturing, focusing on the equipment employed. Other BOP resources are available to describe highly related topics such as cellular assembly considerations, help chain design, team leader roles and responsibilities, work design methodology (standardized work development), and material handling methods.

3.0 PROCESS REQUIREMENTS

Process Summary

Assembly lines will be continuously moving, with slat or belt construction. Successive products will have a defined registration, corresponding to pitch. Process pallets should not be used, but integrating fixtures is a best practice. Assembly conveyor dimensions should be carefully designed to optimize assembly ergonomics. Material presentation will enable removing wasted motion from assembly. All operators will have an andon cord, and a central board will display live production status.

3.1 CONVEYOR EQUIPMENT ATTRIBUTES

3.1.1 CONVEYOR TYPE

Standard: Powered Conveyor

The conveyor type attribute determines how the unit moves throughout the assembly process, and the defined standard is to use a powered conveyor in all manual assembly line operations. No human force should be used to move the product or even to initiate motion of the unit. This solution enables paced cycle time, improved productivity, and reduced variation. Human time is focused on value adding activities.

3.1.2 PACE CONTROL

Standard: Controlled Cycle Time

Pace control sets a consistent work pace for each assembly operation. The standard for this attribute is the length of time a unit is in the operator's pitch should be determined by the conveyor and identical for each work cycle and station. The production unit will remain at each workstation only for the defined line cycle time, leaving the station automatically.

This method enables stable and predictable assembly output because each cycle is identical and variation is minimized. Higher Assembly Yields (AY) can be achieved because problems will be surfaced and solved. Emergency stop and andon systems can be used to stop the line but not to control the cycle length or pace.

Operator controlled work cycles or pace must be avoided. In an indirectly paced cycle, the operator still has the control of the cycle but an aiding device, like a sound or light, is used to minimize cycle variation by indicating ahead or behind conditions. This intermediate step could be used on a path to the defined standard.

3.1.3 UNIT MOTION

Standard: Continuous Movement

Continuous Movement is the defined standard. Continuous movement means the unit constantly moves throughout the operator's work zone (pitch). Continuous movement enables better work methods and achieves the purpose of the assembly conveyor- to bring work to the operator. Continuous movement avoids non-value-adding transfer time within the cycle and uses simpler assembly conveyors, requiring less capital and maintenance than automated indexing conveyors.

A competing method that should be avoided is Stop & Go or Index & Dwell. This refers to any type of line or assembly operation where the unit stops in front of the operator and remains static while the operation takes place, moving afterwards.

3.1.4 PRODUCT FIXTURING ON LINE

Standard: No pallets needed

This attribute refers to the carrier used to move the unit throughout the assembly process. In some cases a pallet is needed for other processes (or shipment) and can be used to aid the assembly process. The defined standard for assembly is: no pallets needed for assembly as the product can be assembled directly on the assembly conveyor.

This attribute is highly related to the product design, and many products will require fixtures due to wheels, etc. being assembled to the bottom of the product. The simplest, best method is to assemble the product directly on a conveyor. If this is not possible, the shipping pallet should be used as an assembly pallet. If this is not possible, and/or practical due to EOL testing, integrated fixtures can be used in place of a dedicated pallet. In this method, the tooling used to secure and convey the product is attached to the assembly line. This can refer to fixture blocks or a turn-table if rotating the product for assembly is absolutely necessary. If a rotating fixture is needed this process should be automated, using the motion of the line.

The least efficient method is the use of a dedicated pallet which needs to be returned to the beginning of the line. This method is even worse when multiple pallet types are needed for each model.

3.1.5 CONVEYOR STYLE

Standard: Slat or belt

This attribute specifies the conveyor type used for assembly lines. The simplest method possible is desired in order to minimize capital and optimize productivity. Specifically, Slat and Belt conveyor types are preferred. Different materials are used in the market, differing in cost, durability,

cleanliness, flexibility, etc. There is not a specified type, since “best” will differ based on specific product needs.

3.1.6 DISTANCE BETWEEN UNITS

Standard: Fixed Pitch

In a continuous moving assembly process, the amount of time for each operation is determined by the spacing of units (pitch) and the speed of the conveyor. If the pitch is not kept constant, the time an operator has to perform standardized work will vary, potentially causing safety, quality, and productivity problems. Every assembly process must ensure repeatable pitch and cycle time per product. Note: pitch is more accurately described as the distance between the same point of successive units rather than width plus distance between because mixed production may result in different space between successive products of different dimensions.

3.1.7 CONVEYOR BED WIDTH

Standard: Narrowest possible

The Bed width attribute refers to how “wide” a conveyor must be, in terms of how far the opposite side of the conveyor is from the assembler. The narrowest possible width (considering product mix) is the best option for any conveyor to minimize reach distance and optimize ergonomics for assemblers. Normally, product depth dictates this dimension, and using a conveyor narrower than the product should be avoided in most cases. Conveyors with minimal or no bed width beyond the slat or belt are preferred.

3.1.8 CONVEYOR BASE

Standard: As Open As Possible

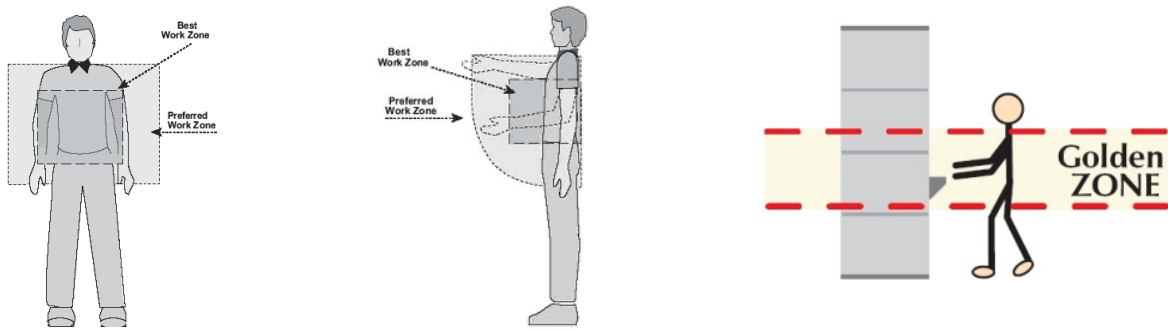
An “open” conveyor base refers to minimizing bed “thickness” and maximizing the open space between the bed and floor. Open space below the bed creates opportunities for material presentation (conveying material under the line) and better ergonomics because the operator will work closer to the product. Space should be provided for operators’ feet under the line regardless of how open the conveyor. Remember, people are our most important asset.

3.1.9 LINE HEIGHT

Standard: Based On Golden Zone

Line Height criteria refers to the distance from the top of the assembly conveyor to the floor. This height is influenced by various factors like product heights and different assembly zones in the same product. Since it is impossible to determine a single recommended line height, the following criteria should be used:

- Position of the product should allow people to be at floor level as much as possible to facilitate material handling, help chain accessibility and reaction speed.
- Position of the product should allow that the operator to work within what is called the strike zone or golden zone to minimize motion and improve ergonomics.
- Differences in working height requirements can be addressed with platforms for operators and elevating materials, always allowing easy access for help chain and response time by not creating barriers. Pits (recesses in the floor) are to be avoided. Elevators for products on the assembly line are to be avoided.



3.1.10 STATIONS SEPARATION

Standard: Group Work, Avoid IPK

The Stations Separation criteria determines how many vacant pitches should be on the assembly line. Empty or vacant workstations should be used sparingly. Adjacent workstations are preferred, but the layout should also ensure easy access for the help chain to respond. The following criteria should be used:

- Stations should be grouped together to minimize line length and response time.
- Stations layout and separation should ensure easy access for help chain.
- Stations at only one side of the line are preferred (see 3.1.13 Assembly Grouping).
- Turns and elevators should be avoided.
- Testing and programming should be in-line and continuous rather than separated from the line.
- New assembly line construction should accommodate future line expansion planning for potential increased demand rather than building in excess (greater than +20%) capacity from the start. Lines should be modular and adjacent floorspace should be free of monuments for this reason.
- IPK or in-process-kanbans can be used for specific intent (enabling TL support or training, etc.), but a vacant pitch for each staffed pitch is not lean or recommended. Use IPK sparingly.

3.1.11 ASSEMBLY LINE BUFFERS

Standard: Avoid Product Accumulation

Assembly line buffers are used to separate assembly sections and accumulate in-process units to absorb recurrent short stoppages. The one-piece flow ideal dictates an assembly line design with minimal lead time and WIP. Buffers should not be used to link two assembly processes but are acceptable to link an automated process with (or within) assembly. An example is the queue between assembly and a bank of DUVs.

3.1.12 ASSEMBLY TAKT TIME

Standard: Develop Takt Time Range Through Trials

Rather than setting a specific requirement for takt time range (and number of assembly lines), the standard is to develop an ideal takt time range through experimentation for each product family. Main considerations of this type of work:

- Elemental cycle work times. Assume part completion, meaning full installation of each component within a single station.

- Assume “ideal” or minimal part presentation. Assembly lines should be optimized for assembly- this means isolating and removing the waste of material handling.
- Test and programming time. This may need to be reduced, broken-up, or duplicated for faster takt times.

Note: More detailed considerations for determining takt time, the number of assembly lines, etc. can be found in the Appendix.

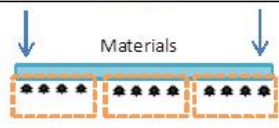
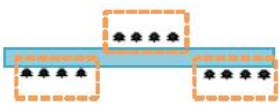
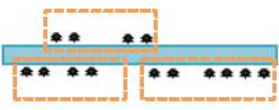
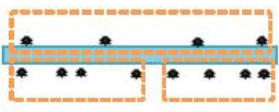
3.1.13 ASSEMBLY GROUPING

Standard: Operators On One Side of Line

Stations should be grouped as close as possible in order to have the shortest process (shorter line) and to minimize help chain response time. Having the operators at one side of the line can create advantages such as reducing material transport and improving help chain response time.

This is a key DFA (Design for Assembly) consideration to avoid unnecessary product turns in the assembly line or unnecessarily long lines in order to have all operators on one side. If the current product design does not allow assembly from only one side, work to avoid isolated stations and group operators as much as possible to minimize help chain response time.

Please refer to the table below as a reference for making decisions on how to group operators.

	Example	Description	Comments
Best		Operators at one side of line	* Faster Help Chain response time (Proper TL Area) * Reduces material transport * Tuggers and material movements in one side * Safer workarea
Better		Stations grouped as close as possible, both sides	* Faster Help Chain response time * Proper TL Area
Good		Stations grouped as close as possible, both sides	* Larger TL area * Materials in both sides of the line
Bad		Isolated Stations, large TL area	* Larger number of workstations and a larger area for TL (Longer response times) * TL in both sides (Help Chain more difficult) * Materials in both sides of the line * Inefficient Material transport

3.1.14 FUNCTIONAL TEST SYSTEM

Standard: On-line Continuous

On-Line Continuous: testing is performed at the assembly line flow. Off-line static: Units are taken away from assembly lines for testing to be performed. The testing process should be on-line continuous.

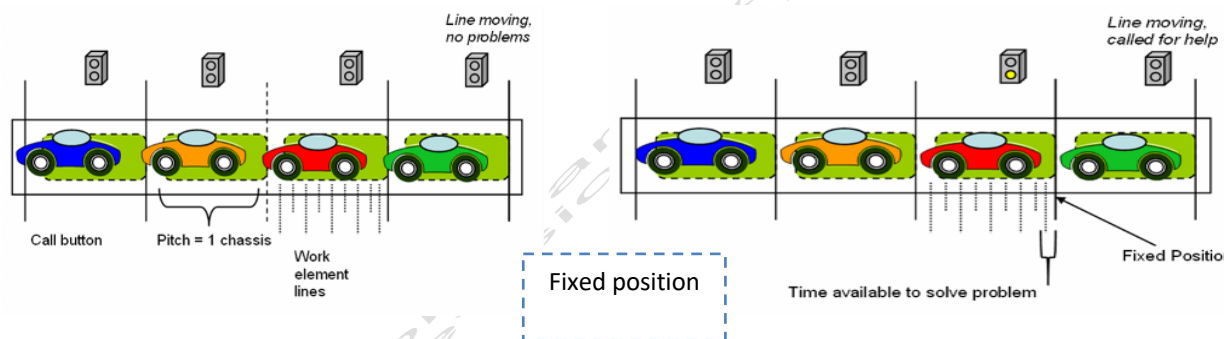
3.1.15 EQUIPMENT SUPPORT STRUCTURE

Standard: Overhead, Suspended

This attribute refers to how assembly equipment such as part manipulators, air hoses, scanners, task lighting, etc. are supported overhead. The structure used for utilities installation should be suspended from the ceiling, if possible, in order to avoid any visual obstruction for Team Leader or physical obstruction for material presentation. If ceiling support is not possible, a cantilevered structure is preferred to minimize physical obstructions. Support column location should be carefully planned to minimize obstruction and maximize flexibility in improving assembly and material handling/presentation operations.

3.1.16 FIXED POSITION STOPS**Standard: Andon Line Stops Only At Fixed Position Stops**

The assembly conveyor should allow for a Fixed Position Stop (FPS). FPS means that when an andon is triggered, the line won't stop until reaching a predetermined position, finishing its cycle. Instant stop, meaning that line movement stops as soon as the andon is triggered is not recommended. Allowing the work cycle to continue enables time for the Help Chain (TL) to react and assess the problem without stopping the line and interrupting production of all linked operations on the line. Cycle work won't be interrupted as often while fulfilling the requisite of not forwarding problems to the next subsequent process. A fixed position stop allows every operator to finish current cycle and assemble all parts that were in process.

**3.1.17 SEQUENCE****Standard: Ensure Sequence**

All assembly lines should ensure sequence. Assembly lines which do not maintain sequence throughout the line create unnecessary barriers to efficient material handling and Standardized Work. In addition, these assembly lines hide problems and often create large differences in machine utilization. All assembly lines should be designed according to the lean ideal of supporting dedicated, predictable flow paths for all products.

The best practice is to only have one offline repair area at the end of the assembly line. Captive (within the assembly line) capital equipment should ensure FIFO (First In, First Out). The first products entering the assembly line are first ones to exit the process. In other words, the products are being produced in the same order they are started.

3.1.18 PROCESS LOOPS**Standard: No Process Loops**

Process loops should not be used on any assembly line. This type of conveyor or off-line handling (on carts, etc.) of units is sometimes used following a repair operation, looping the product back through part of the assembly process (often to validate the repair through testing). Repair loops

not only disrupt sequence, but they also waste assembly line capacity and degrade assembly yield. A best practice is to add an additional testing equipment as part of the repair area.

3.1.19 SUBASSEMBLIES

Standard: Coupled

Sub-assemblies, work in parallel of the main assembly line, need to be coupled to the main conveyor. Coupled subassemblies feed the line one-to-one with minimal inventory. A small buffer may be added to compensate for variation. Subassemblies should be located as close to the line as possible. Coupled subassemblies do not require scheduling, minimize WIP, and encourage faster response to problems. Co-location eliminates isolation and allows TLs to support the subassembly.

3.2 MANUFACTURING EXECUTION SYSTEM ATTRIBUTES

3.2.1 ASSEMBLY LINE PRODUCTION DATA

Standard: Fully Automatic

All assembly line data collection should be fully automated, avoiding manual entry of the metrics/numbers to a computer or board. Production counts, repairs, andon signals, etc. should be collected by scanners, PLC counters, or other automatic devices. A MES (Manufacturing Execution System) should record, calculate, and report, and retain the following:

- Andon counts by station
- Andon downtime by station
- Other downtime events- stop buttons engaged, line blocked downstream, equipment downtime, etc.
- Assembly Yield (see 3.2.2 below)
- Number of defects and First Pass Yield.

3.2.2 REAL TIME AY DISPLAY

Standard: Display Real Time Production Progress

Assembly Yield (AY) should be presented as real time progress versus real time targets. Displaying progress compared to hourly, shift, or day targets during any of those respective periods is not sufficient. Additionally, the lost units should be compared to the total possible so boards would always show the highest possible AY that the shift will be able to achieve based on the units lost at that point in time.

3.2.3 DEFECT DETECTION AT STATIONS

Standard: Automated Critical Quality Validation

Confirmation and feedback of critical to quality checks should be completed automatically during the work cycle with no additional time and elements required for inspection or confirmation. Manual quality confirmation elements should only be used as a short term countermeasure while a long term solution is developed. Examples of automatic validation processes include fastener torque monitoring/validating/collecting systems, non-contact measurement, or part presence validation (via machine vision). These quality verification solutions should not require operator cycle work but should provide visual or audible confirmation feedback.

3.2.4 NO DEFECT FORWARDING

Standard: No Defect Forwarding

Confirmation of critical to quality checks which identify nonconformances must ensure no defects are forwarded to the customer. Work elements which can be corrected within the station (part presence, fastener torque, etc.) should be completed and confirmed prior to allowing the product to advance to the next workstation. The MES system should verify “pass” (not just absence of failure) of critical quality inspections prior to allowing the product to advance past the FPS (fixed position stop, described in 3.1.16). Work elements which cannot be corrected within the workstation can be forwarded to the assembly line repair area, but the MES should require team leader, test operator, or repair operator approval to route a product to repair. Also, the MES should lock any “failed” product for shipment until a “pass” has been achieved for the failed check.

3.2.5 WORKSTATION DISPLAY***Standard: Display Work Instructions***

A best practice is to leverage MES systems to display critical information at the assembly process. This is especially important if an assembly line runs mixed production of multiple models. In these cases the MES will leverage traceability data to retrieve and display product specific information. Typical application use cases for display are work instructions, single point lessons (SPL), and changeover instructions.

Work Instructions (WI)- WI are typically controlled documents based on pictures of the stage of assembly being completed with call-outs for critical work elements and sequence of assembly. WI display is helpful for operator training, process confirmation, and as a reference for the Help Chain. This WI display is not expected to be a step-by-step assembly instruction referenced by the assembly operator every cycle, but it is helpful when an operator returns to a workstation after a period of time or a model changeover starts. It is also important to note that Work Instructions are not a substitute for Standardized Work development and improvement. Standardized Work attributes are defined in a separate document. Work Instructions are also not a substitute for training. A trained assembler will not need to refer to this type of document every cycle, and work design should not allow time for constant reference.

Single Point Lessons (SPL)- A SPL is used to visualize critical assembly steps, at a very detailed level. This tool is often used in response to quality problems or to call special attention to assembly elements that can be mis-assembled and cannot be mistake-proofed. Information in SPL should be 80%+ visual and focused on the “how” of elemental work completion.

Changeover Instructions (C/O)- Assembly lines running batch production require material and equipment changes. A best practice is to use the MES to drive the C/O instructions at each workstation. Note: in 1X1 and mixed production systems MES can drive manifest information at each workstation, which is effectively a continual C/O instruction.

3.3 MATERIALS ATTRIBUTES**3.3.1 MATERIALS DELIVERY- PHYSICAL SEPARATION*****Standard: Physical Separation***

Assembly of complex and large electromechanical devices requires deliver of heavy parts to the assembly line, and medium to high production volumes creates sufficient “traffic” of material handling to create safety risk for those working in the assembly areas. In order to keep our

associates safe, all assembly operations should be physically separated from powered industrial vehicles (PIV).

3.3.2 MATERIALS LOCATION

Standard: Front of Operator

Assembly materials location is important because it directly impacts ergonomics and productivity of each operation and the entire assembly process. A common workstation design locates material behind the operator. This concept has the advantage of separating material delivery from assembly operator, but this location is sub-optimal because of the operator movement and rotation required to pick the materials to be assembled. Locating parts at one side of the operator can improve this condition (reducing movement and rotation), but this location also creates a barrier impacting the Help Chain. Material should be located in Front of the Operator if possible (the larger the product, the more difficult to comply with this standard). Material can be presented next to, above, or below the product, eliminating the need to turn to pick material. This implementation is usually achieved through several, iterative improvement efforts.

3.3.3 MATERIALS PRESENTATION

Standard: Point of Assembly

The assembly materials presentation varies greatly in most assembly operations. The defined best method to present material to the operator is exactly at the point of assembly, in the correct orientation, minimizing movement to position and secure. This ideal is also the most difficult to design, and requires more off-line standardized work development. Several other methods are typical in our assembly processes. A list is presented here from worst to best:

Bulk Containers: Pallet-sized, ship-quantity containers are brought to the assembly process with forklifts or tuggers. These containers consume valuable floorspace at the assembly process, often require handling or break-down of large packaging, and should be avoided.

Line-side Flowrack: To avoid bulk containers, a custom rack is designed to present the parts needed for the operation. Rules are developed for length, WIP levels, etc. This eliminates some problems associated with the bulk containers (such as removing longer periodic work) but is not the optimal solution. Workstations with parts that vary from product to product may require more space for part presentation than available within pitch.

Single Pick Point: A kit is assembled near line-side to eliminate the fluctuation of having many different models of the same part type in a rack or to remove periodic work from the assembly operator. Racks are decreased in size and fluctuation is transferred to the material handlers.

Exact Point of use: Parts to be assembled are presented at the point of assembly, in front of the operator, in the appropriate orientation, and with the minimal distance from the material to the physical location where the material is going to be assembled. This solution is the defined standard for assembly processes.

Note: the ideal described above is typically achieved using a Creform (or similar) custom flowrack to present the parts. Another innovation that has created new and better opportunities for material presentation is the use of a collaborative robot to present materials to point of assembly.

3.3.4 MATERIALS DELIVERY METHOD

Standard: Automated Delivery

This attribute is similar to the MATERIALS PRESENTATION attribute above as there is a progression of worst to better. The defined standard is to automate the delivery of parts to assembly with Automated Guided Vehicles (AGV). Other material delivery options often transition through the following progression:

1. Bulk materials are brought to assembly with a forklift.
2. The obvious waste in handling bulk containers drives usage of lineside flowracks to improve part presentation to the operator.
3. Flowracks enable and encourage tugger delivery to drive efficiency in material delivery and floorspace.
4. Regular material delivery routes are automated with an AGV which follows a repeating path while lineside material handlers (water spiders) pick delivered parts fill the lineside racks.
5. Specialized carts are designed to be delivered by AGV and mated directly to lineside racks, eliminating secondary (water spider) handling of the parts.
6. Ideally, carts are sourced from suppliers pre-loaded and ready to deliver to lineside.

Note: off-line material handling work needs the same detailed study and improvement as on-line assembly work. Material handlers should have cyclical standardized work defined, and the work must be analyzed and improved to increase productivity using the same methods as assembly labor improvements. This work cannot be skipped. All steps targeted for automation should be developed as manual processes first then improved before automating.

3.4 CONVEYOR EQUIPMENT ATTRIBUTES**ANDON SYSTEMS**

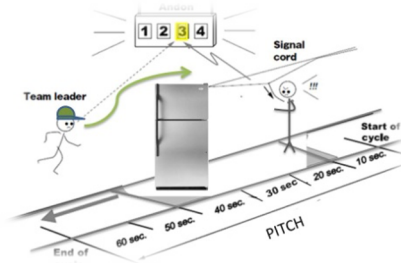
An andon system is used to notify the help chain (team leaders, management, maintenance, etc.) of a quality or process problem. The alert can be activated manually by a worker or may be activated automatically by the production equipment itself. The system should include a means to stop production so the issue can be corrected.

“An Andon system is one of the principal elements of the Jidoka quality-control method pioneered by Toyota as part of the Toyota Production System. It gives the worker the ability to stop production when a defect is found, and immediately call for assistance. Common reasons for manual activation of the Andon are part shortage, defect created or found, tool malfunction, or the existence of a safety problem. Work is stopped until a solution has been found although may not implemented yet”⁴.

⁴ Liker, Jeffrey (2004) "The Toyota Way" New York: McGraw Hill ISBN 0-07-139231-9

3.4.1 ANDON SWITCHES QUANTITY***Standard: Andon Per Station***

Every operator must be able to call for help immediately. Minimal operator motion and effort should be used in order to activate the andon switch. The assembly process needs an andon per workstation so each station is equipped with a triggering device to call for help.



3.4.2 ANDON SWITCH TYPE

Standard: Cord

The andon switch should be simple, easy to activate, and easily reached. It also needs to be clear of the working area and allow the Team Leader easy access. Since the assembly areas are constantly changing, it also needs to be easy to move. The defined standard for Andon Switch is a cord activated switch. The cord could be mounted horizontally above the work area or hanging vertically depending on the system but needs to be easily accessed. Best examples of andon switches are inexpensive, do not interfere with the work station, and are easy to move. Wireless andon options with cords that can be moved around the assembly area and re-located as necessary are optimal.

Buttons, either hanging or fixed to a structure should not be used. Buttons are harder to activate and usually are larger than a cord, interfering more with the work area. The least preferred solution is a fixed button, set on a fixed structure.

3.4.3 ANDON SYSTEM LIGHTS

Standard: Light Per Station

An andon system light is used to initiate the help chain, like Team Leaders, to easily locate where a problem has occurred. The main purpose is to initiate response as fast as possible. In order to locate the problems faster, an andon light per station is required. This means that each station is equipped with a light. The light should be easily seen by lineside support. Note- smaller lines (<20 stations) may only use an andon board rather than lights at each station.

3.4.4 ANDON SYSTEM AUDIO

Standard: Audio

Audible andon is required in the assembly processes as part of the notification system. An audible andon is a sound or melody that starts when an andon switch is triggered, notifying Team Leader and the Help Chain that some out of standard events took place and that an operator needs help. The alert should continue until a TL responds and clears the andon.

3.4.5 ANDON BOARDS

Standard: Andon Board Displays Status For Each Assembly Line

An Andon board is a display that continuously shows changing status of the production line, and one Andon Board per line is needed. This board shows which station is having a problem and should be visible from the whole assembly line. A monitor or projector is recommended over "scoreboard" type boards due to flexibility.

3.4.6 ANDON BOARDS INFORMATION

Standard: Display Station Status, Production Information

Electronic andon board should include station status as well as Production (AY) & Quality (FPY) information. The andon board should indicate the section that is stopped or having problems as well as latest production and quality accomplishments.

4.0 PRODUCT DESIGN REQUIREMENTS

Product design has a major impact in assembly performance. Product design determines assembly complexity in part numbers and also in assembly methods. A comprehensive guideline for product design requirements is beyond the scope of this document, but several key points are listed below:

- The Advanced Manufacturing Engineering team must be engaged in the product design process as early as possible to ensure Design for Assembly (DFA) considerations are incorporated. Manufacturing input to the product development process needs to communicate clear and aligned requirements from Assembly operations even prior to Alpha builds.
- The Assembly Operations team should maintain product line specific requirements and defined process needs, communicated to the hardware design team as on-going continuous improvement and in preparation of successive product generations. Best practices should be capture and communicated.
- In addition to minimizing the number of pieces needed for a product and facilitating assembly, the design must eliminate complexity and variation between products. Modularity in product design is a key enabler to improving assembly operations.
- The product design should support Assembly from a single side of the line as much as possible.
- Design for Automated assembly is outside the scope of this document, and requires an entirely different set of attributes.
- Design for Assembly (DFA) principles are captured in a companion document.

5.0 APPENDIX

DETERMINING CONVEYOR SPEED and TAKT TIME

Takt Time and Cycle Time

The terms “cycle time” and “takt time” are often used interchangeably in assembly operations. Cycle time is the total amount of work time at each station for all assembly work (cycle work + distributed periodic work) while takt time is the rate of customer demand calculated by dividing the amount of time available for work by the amount of customer demand during that period (typically expressed as seconds/unit). Each assembly line is set to an overall process cycle time which establishes a work balance limit for each operation. In a perfect world cycle time of the line equals takt time. However, problems that degrade Assembly Yield as well as factors such as using multiple processes, building external buffers, or making to stock result in differences between true takt time and cycle time of the line. Assembly line production rates are normally referred to as “takt time” even though this is not technically correct. The more important point is that assembly processes and all upstream feeders must be designed to create “takt image” for employees responsible for completing cycle work. “Takt time” is often used in this document and

others to refer to the work balance time available for assembly work or the time for an assembly line to cycle one pitch. Larger pitches result in longer lines and faster conveyor speeds for a given takt time.

Determining Takt Time

Many considerations are involved in deciding which Takt Time is best for assembly of a given product. Customer demand, inventory levels, and shipment forecasts are all critical information to understand or estimate when determining Takt Time. However, it is not typically as simple as setting Takt Time to match customer demand for a given product family. For higher volume product families, there may be the option to divide the volume across multiple assembly lines which could also run at different rates. For lower volume product families, there may be the option to combine the product volume with that of other similar products on a single, faster line.

Some comparison data can be used from similar processes in other industries. Automotive assembly (final vehicle as well as subassembly lines) takt times are typically designed to approximately one minute. Appliance assembly lines are typically faster than one minute (~30 seconds), with only rare exceptions above 60 seconds. This is true for all appliance types with smaller (microwaves) products typically being faster (20-30) and larger (refrigerators) product lines designed for slower (30-40s) takt times.

Some advantages of faster takt times:

- Higher productivity, less takt gap (insaturation)
- More options for balancing work
- Less training time per workstation
- Less mental load on operators
- More opportunities to catch errors in process
- Better capital investment efficiency
- Better floorspace efficiency (minimizes number of lines)
- Better material presentation (fewer parts/workstation)

Some advantages of slower takt times

- Easier to design work for part completion (all cycle work for a given component in one workstation)
- More time to respond to andon within work cycle
- Easier to accommodate test and programming cycles without duplicating or dividing
- Lines are smaller and easier to manage (though waste is more easily hidden)

6.0 GLOSSARY

AME: Advanced Manufacturing Engineering

Andon: Manufacturing term referring to a system to notify management, maintenance, and other workers of a quality or process problem

AY: Assembly Yield. This is the main productivity metric for assembly lines. AY is calculated by dividing the actual quantity produced by the assembly process by the target production quantity for a given period of time. Maximum AY is 100% due to fixed pitch and continuous assembly

conveyors. If an assembly conveyor runs all day with no stoppages for unplanned downtime, AY should be 100%. AY is not degraded with FPY, these are independent measures.

FIFO: First In, First Out. The first products entering the assembly line are first ones to finish the process.

FPY: First Pass Yield. FPY measures the percentage of products produced without requiring repair area intervention. FPY does not capture on-line corrections.

SW: Standardized Work is the most efficient way to do work, centered around human movements. The best way for making quality products in the required time, safely, with minimum waste.

TL: Team Leader is an hourly operator who supports the operations and solves problems. This person is the first to respond to andon. The Team Leader is not the team's supervisor. There are resources available for TL roles and responsibilities in another Bill of Process.

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REVISION HISTORY

Rev.	ECO #	Description of Change	Sections Affected	Originator/Department
01	ECO-018033	Initial Release	All	AME Mgr – Paul Seay